ASEN-6062

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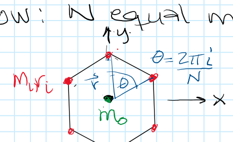
HWK #2

8) given: central configuration

$\rho(\underline{X}) = U^2 I_p \qquad \frac{\partial \rho}{\partial \underline{X}} = \vec{0}_{3N\times 1}$

- show: N equal masses @ polygon vertices are cc



& evaluate $\sigma \qquad r = r_i \quad m = m_i \qquad \sigma = \frac{U}{2I_p}$

$U = -\frac{1}{2} G \sum_{i \neq j}^{N} \frac{m_i m_j}{|\vec{r}_{ij}|}$

$I_p = \sum_{i=1}^{N} m_i r_i^2 = N m r^2 \qquad \sigma = \frac{U}{2I_p}$

$\underline{X} = \begin{bmatrix} \vec{r}_1 \\ \vec{r}_2 \\ \vdots \\ \vec{r}_N \end{bmatrix} \qquad \frac{\partial I_p}{\partial r_i} = 2\sum_{i=1}^{N} m_i r_i = 2Nmr$

$|\vec{r}_{ij}| = |\vec{r}_i - \vec{r}_j| = r \sin\frac{2\pi(i-j)}{N} = 2r \sin\frac{\pi(i-j)}{N}$

$U = -G \sum_{i \neq j}^{N} \frac{m_i m_j}{2r\sin\frac{\pi(i-j)}{N}} = -\frac{Gm^2}{2r} \sum_{1\le k \ne j \le N} \frac{1}{\sin\frac{\pi(i-j)}{N}}$

$\sigma = \dfrac{-\frac{NGm^2}{2r} \sum_{k=1}^{N-1} \frac{1}{\sin\frac{\pi k}{N}}}{2Nmr^2}$

$\boxed{\sigma = \frac{Gm}{4r^3} \sum_{k=1}^{N-1} \csc\frac{\pi k}{N}}$

Euler's: $\frac{\partial \rho}{\partial \underline{X}} = 0$

$\rho = U^2 I_p = \left[\frac{NGm^2}{2r} \sum_{k=1}^{N-1} \csc\frac{\pi k}{N}\right]^2 N m \cancel{r^2}$

$\rho = \frac{N^3 G^2 m^3}{2} \left[\sum_{k=1}^{N-1} \csc\frac{\pi k}{N}\right]^2$ Not dependent on r

$\boxed{\frac{\partial \rho}{\partial \underline{X}} = 0}$

- show: can add mass at center and keep cc

evaluate σ

$I_p = m_0 \cdot 0 + \sum_{i=1}^{N} m_i r_i^2 = N m r^2$

$U_{com} = G\sum \frac{m_0 m_i}{r} = \frac{GNm_0 m}{r}$

$U = -\frac{GNm_0 m}{r} + \frac{NGm^2}{4r} \sum_{k=1}^{N-1} \csc\frac{\pi k}{N}$

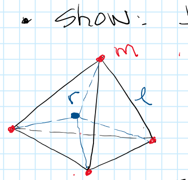
$\sigma = \frac{U}{2I_p} = \dfrac{\frac{GNm}{r}\left(m_0 + \frac{m}{4} \sum_{k=1}^{N-1} \csc\frac{\pi k}{N}\right)}{Nmr^2}$

$\boxed{\sigma = \frac{G}{r^3}\left(m_0 + \frac{m}{4} \sum_{k=1}^{N-1} \csc\frac{\pi k}{N}\right)}$

$\rho = U^2 I_p = \left[\frac{GNm}{\cancel{r}}\left(m_0 + \frac{m}{4} \sum_{k=1}^{N-1} \csc\frac{\pi k}{N}\right)\right]^2 N m \cancel{r^2}$

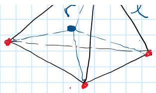
$\boxed{\frac{\partial \rho}{\partial \underline{X}} = 0}$

- show: tetrahedron is cc



$r = l\sqrt{\frac{3}{8}} \qquad l = \sqrt{\frac{8}{3}}\, r$

$I_p = \sum_{i=1}^{4} m_i r_i^2 = 4mr^2$



$$I_p = \sum_{i=1}^{4} m_i r_i^2 = 4mr^2$$

$$X = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_N \end{bmatrix}$$

$$\mathcal{U} = -G \sum_{i=1}^{3} \sum_{\substack{j=1 \\ i \neq j}}^{4} \frac{m^2}{l} = -G\left[\frac{m^2}{r_{12}} + \frac{m^2}{r_{13}} + \frac{m^2}{r_{14}} + \frac{m^2}{r_{23}} + \frac{m^2}{r_{24}} + \frac{m^2}{r_{34}}\right] \qquad r_{12} = r_{13} = \dots = l$$

$$\mathcal{U} = -\frac{6Gm^2}{l} = -\frac{6Gm^2}{\sqrt{8/3}\, r}$$

$$\sigma = \frac{\frac{6Gm^2\sqrt{3}}{\sqrt{8}\, r}}{4mr^2} = \frac{3}{2}\frac{Gm\sqrt{3}}{\sqrt{8}\, r}$$

$$\boxed{\sigma = \frac{3\sqrt{3}\, Gm}{4\sqrt{2}\, r^3}}$$

$$\rho = \left[\frac{6Gm^2}{\sqrt{8/3}\, \cancel{r}}\right]^2 4m\cancel{r^2}$$

$$\boxed{\frac{\partial \rho}{\partial X} = 0}$$

9) given: planar central config w/ homographic soln

$$\ddot{\alpha} - \alpha\Omega^2 = \frac{2\sigma^*}{\alpha^2} \qquad \sigma^* < 0 \qquad \beta = \alpha^2\Omega = 0 = \text{total ang momentum}$$

$$\alpha\dot{\Omega} - 2\dot{\alpha}\Omega = 0 \qquad \sigma^* = \frac{1}{2}\frac{\mathcal{U}}{I_p} \qquad \alpha = \text{scaling} \quad \Omega = \text{rotation rate}$$

discuss: solutions for +, −, 0 total energy

$$\ddot{\alpha} - \frac{\beta^2}{\alpha^3} = \frac{2\sigma^*}{\alpha^2}$$

if $\beta = 0$ then $\alpha = 0$ (collision) or $\Omega = 0$ (not rotating)

$$\ddot{\alpha} = \frac{2\sigma^*}{\alpha^2}$$

$\Rightarrow$ radial motion only

$$\int \dot{\alpha}\ddot{\alpha} - \int \frac{2\sigma^*}{\alpha^2}\dot{\alpha} = 0$$

$$\frac{\dot{\alpha}^2}{2} = E - \frac{2\sigma^*}{\alpha}$$

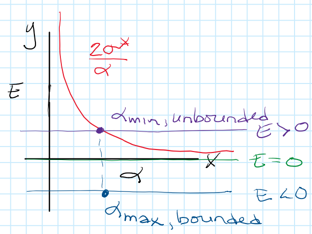
$E > \frac{2\sigma^*}{\alpha}$ always

$$\frac{\dot{\alpha}^2}{2} + \frac{2\sigma^*}{\alpha} = 0$$

$$E = \underbrace{\frac{\dot{\alpha}^2}{2}}_{KE} + \underbrace{\frac{2\sigma^*}{\alpha}}_{\text{potential}} \qquad E = +, -, 0$$

$$\frac{\dot{\alpha}^2}{2} = E - \frac{2\sigma^*}{\alpha}$$

since $\frac{\dot{\alpha}^2}{2} > 0$ $\quad E > \frac{2\sigma^*}{\alpha}$



$E = 0$: (parabolic)

$$\frac{\dot{\alpha}^2}{2} = \frac{2\sigma^*}{\alpha} \qquad \alpha \to \infty^+ \quad \dot{\alpha} \to 0$$

$$\alpha \to 0 \quad \dot{\alpha} \to \infty$$

$E > 0$: (hyperbolic)

$$E = \frac{\dot{\alpha}^2}{2} + \frac{2\sigma^*}{\alpha} > 0$$

$$E - \frac{\dot{\alpha}^2}{2} > \frac{2\sigma^*}{\alpha}$$

$$\alpha > \frac{2\sigma^*}{E - \frac{\dot{\alpha}^2}{2}}$$

$$\alpha > \frac{2\sigma^*}{\frac{2E - \dot{\alpha}^2}{2}} > \frac{\sigma^*}{2E - \dot{\alpha}^2}$$

@ $\dot{\alpha} = 0$ $\quad \alpha > \frac{\sigma^*}{2E}$ $\rightarrow$ minimum α

$$\alpha \to \infty \qquad \frac{2\sigma^*}{\alpha} \to 0 \qquad \frac{\dot{\alpha}^2}{2} \to \infty$$

$E < 0$: (elliptical)

$$E = \frac{\dot{\alpha}^2}{2} + \frac{2\sigma^*}{\alpha} < 0$$

$$\alpha < \frac{\sigma^*}{2E - \dot{\alpha}^2}$$

@ $\dot{\alpha} = 0$ $\quad \alpha < \frac{\sigma^*}{2E}$ $\rightarrow$ maximum α

$\alpha \to \infty$ $\quad \frac{2\sigma^*}{\alpha} \to 0 \Rightarrow \frac{\dot{\alpha}^2}{2} < 0$ X Not possible $\Rightarrow$ bounded

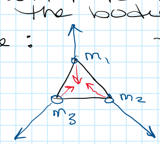
$\alpha \to 0^+$ $\quad \frac{2\sigma^*}{\alpha} \to \infty$ $\quad \frac{\dot{\alpha}^2}{2} \to \infty$ X Not possible

$\alpha \to \infty \quad \frac{2\alpha^*}{\alpha} \to 0 \Rightarrow \frac{\dot{\alpha}^2}{2} < 0$ X Not possible $\Rightarrow$ bounded

$\alpha \to 0^+ \quad \frac{2\alpha^*}{\alpha} \to \infty \quad \frac{\dot{\alpha}^2}{2} \to \infty$ X Not possible

Discussion: Since there is no angular momentum the rotation rate = 0 & there is only radial motion of the bodies from the attracting body

for example: the configuration is expanding or collapsing with rectilinear motion.



For the case where E = 0, this is equivalent to a parabolic orbit where the bodies escape but velocity decreases as the distance increases, eventually going to 0 at infinite distance. As the distance decreases, the velocity of the bodies increase & result in total collapse.

For the case where E > 0, which is equivalent to a hyperbolic orbit, there is a minimum distance that the bodies can achieve $\left(\frac{\alpha^*}{2E}\right)$. This distance is determined by velocity going to zero. As the distance increases velocity increases to infinity & is unbounded. If the body comes in from ∞, it will reach $\frac{\alpha^*}{2E}$, rebound & continue to infinity.

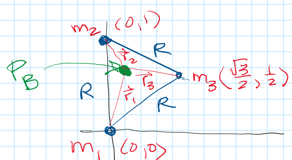
The E < 0 case is bounded with a maximum distance of $\frac{\alpha^*}{2E}$. The bodies' velocity slows to near zero & the configuration expands to this point then it collapses to $\alpha = 0$ (collision).

10) given: 3BP Lagrange config

$m_i \geq 0 \quad m_1 + m_2 + m_3 = 1 \qquad \Sigma m_i = 1$

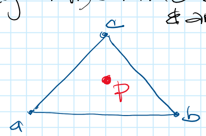
show: COM can lie within or on boundary depending on relative masses

$R = r_{12} = r_{13} = r_{23}$



using barycentric coordinates & area ratios

$P = \alpha a + \beta b + \gamma c$



$\alpha, \beta, \gamma \geq 0$
$\alpha + \beta + \gamma = 1$

$R_c = m_1\vec{r}_1 + m_2\vec{r}_2 + m_3\vec{r}_3$

$R_{c_x} = m_1(0) + m_2(0) + m_3\left(\frac{\sqrt{3}}{2}\right)$

$R_{c_y} = m_1(0) + m_2(1) + m_3\left(\frac{1}{2}\right)$

$R_c = \left(\frac{\sqrt{3}}{2} m_3,\ m_2 + \frac{m_3}{2}\right)$

$A_{tot} = \frac{1}{2}\det\begin{pmatrix} r_{2x}-r_{1x} & r_{3x}-r_{1x} \\ r_{2y}-r_{1y} & r_{3y}-r_{1y}\end{pmatrix} = \frac{1}{2}\det\begin{pmatrix} 0 & \frac{\sqrt{3}}{2} \\ 1 & \frac{1}{2}\end{pmatrix} = -\frac{\sqrt{3}}{4}$

replace $\vec{r}_1$ w/ P

$A_1 = \frac{1}{2}\det\begin{pmatrix} 0-P_x & \frac{\sqrt{3}}{2}-P_x \\ 1-P_y & \frac{1}{2}-P_y\end{pmatrix}$

$= \frac{1}{2}\left[-P_x\left(\frac{1}{2}-P_y\right) - (1-P_y)\left(\frac{\sqrt{3}}{2}-P_x\right)\right]$

$= \frac{1}{2}\left(\frac{P_x}{2} + \cancel{P_xP_y} - \frac{\sqrt{3}}{2} + P_x + \frac{\sqrt{3}}{2}P_y - \cancel{P_xP_y}\right)$

$= \frac{1}{2}\left[-\frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2}P_y + \frac{P_x}{2}\right]$

$\alpha = \frac{A_1}{A_{tot}} = \frac{-\frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4}P_y + \frac{P_x}{4}}{-\frac{\sqrt{3}}{4}} = 1 - P_y - \frac{P_x}{\sqrt{3}}$

$A_2 = \frac{1}{2}\det\begin{pmatrix} P_x & \sqrt{3}/2 \\ P_y & 1/2\end{pmatrix} = \frac{P_x}{4} - \frac{\sqrt{3}}{4}P_y$

$\beta = \frac{\frac{P_x}{4} - \frac{\sqrt{3}}{4}P_y}{} = -\frac{P_x}{\sqrt{3}} + P_y$

$$A_2 = \frac{1}{2}\det\begin{pmatrix} P_x & \sqrt{3}/2 \\ P_y & 1/2 \end{pmatrix} = \frac{P_x}{4} - \frac{\sqrt{3}}{4}P_y$$

$$\beta = \frac{\frac{P_x}{4} - \frac{\sqrt{3}}{4}P_y}{-\frac{\sqrt{3}}{4}} = -\frac{P_x}{\sqrt{3}} + P_y$$

$$A_3 = \frac{1}{2}\det\begin{pmatrix} 0 & P_x \\ 1 & P_y \end{pmatrix} = -\frac{1}{2}P_x$$

$$\gamma = \frac{-\frac{P_x}{2}}{-\frac{\sqrt{3}}{\cancel{4}2}} = \frac{2P_x}{\sqrt{3}}$$

$$\alpha + \beta + \gamma = 1 - \cancel{P_y} - \cancel{\frac{P_x}{\sqrt{3}}} - \cancel{\frac{P_x}{\sqrt{3}}} + \cancel{P_y} + \cancel{\frac{2P_x}{\sqrt{3}}} = 1 \checkmark$$

$$R_{c_x} = \frac{\sqrt{3}}{2}\gamma = \frac{\sqrt{3}}{2}\left(\frac{2}{\sqrt{3}}\right)P_x = P_x \checkmark$$

$$R_{c_y} = \beta + 1/2\,\gamma = -\frac{P_x}{\sqrt{3}} + P_y + \frac{P_x}{\sqrt{3}} = P_y \checkmark$$

Check some scenarios:

$m_1 = m_3 = 0 \quad m_2 = 1$

$R_c = \vec{r}_2 \Rightarrow$ on vertex

$m_1 = m_2 = m_3 = \frac{1}{3}$

$R_c = \frac{1}{3}\vec{r}_1 + \frac{1}{3}\vec{r}_2 + \frac{1}{3}\vec{r}_3 \Rightarrow$ at centroid

$m_1 = m_2 = \frac{1}{2} \quad m_3 = 0$

$R_c = \frac{1}{2}\vec{r}_1 + \frac{1}{2}\vec{r}_2 = \frac{\vec{r}_1 + \vec{r}_2}{2} \Rightarrow$ on edge

Any point is inside triangle if:

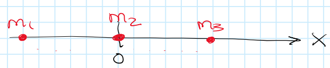
$0 < \alpha < 1$

$0 < \beta < 1$

$0 < \gamma < 1$

11) given: Euler co linear central configurations $m_1 = m_2 = m_3$

find: all solutions



$$\frac{\partial U}{\partial \vec{r}_i} + \frac{U}{2I_p}\frac{\partial I_p}{\partial \vec{r}_i} = 0$$

$$\frac{\partial U}{\partial \vec{r}_i} = \frac{U}{2I_p}\frac{\partial I_p}{\partial \vec{r}_i}$$

$x_1 = -r_{12} \quad x_2 = 0 \quad x_3 = r_{23}$

$x_1 + x_2 + x_3 = 0$ com @ origin due to symmetry

$r_{12} = |x_2 - x_1| \quad r_{23} = |x_3 - x_2| \quad r_{13} = r_{12} + r_{23}$

$U = \sum_{1 \le i < j \le N} \frac{m_i m_j}{r_{ij}}$ $\quad I_p = \sum_{i=1}^{N} m_i r_i^2$

$U = \frac{m^2}{r_{12}} + \frac{m^2}{r_{23}} + \frac{m^2}{r_{13}}$ $\quad I_p = m x_1^2 + m x_2^2 + m x_3^2$

$I_p = m r_{12}^2 + m r_{23}^2$

@ x_2

$\frac{\partial U}{\partial x_2} = \frac{U}{2I_p}\frac{\partial I}{\partial x_2}$ $\quad \frac{\partial I}{\partial x_2} = 2 m x_2$

$$\frac{\partial U}{\partial x_2} = -\frac{m^2}{r_{12}^2}(1) + \frac{m^2}{r_{23}^2}(-1) +$$

$$\frac{\partial U}{\partial x_2} = m^2\left(-\frac{1}{r_{12}^2} + \frac{1}{r_{23}^2}\right) = \frac{U}{2I_p}(2m\cancel{x_2})^0$$

$$\frac{1}{r_{12}^2} = \frac{1}{r_{23}^2}$$

$\boxed{r_{12} = r_{23}}$ $\quad r_{13} = 2r_{12}$

$r = r_{12} = r_{23} \quad r_{13} = 2r$

$\boxed{\text{all masses equidistant from each other}}$ there is only one solution

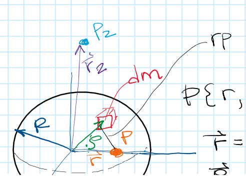
12) given: $\sigma = \text{const}$ sphere radius R mass distribution

particle p @ r distance from center

$M = \frac{4\pi}{3}\sigma R^3$

$dU = -\frac{G\,dm}{|r - \rho|}$ $\quad \rho$ = location of dm

$dm = \sigma dV$



$P\{r, \theta, \phi\}$

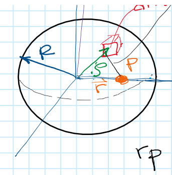
$\vec{r} = (r, 0, 0)$

$M = \frac{4\pi}{3}\sigma R^3$

$dU = -\frac{G\,dm}{|r-\rho|}$ ρ = location of dm

$dm = \sigma\,dV$

$\sigma(\vec{r}) = \begin{cases} 0 & r \notin B \\ \neq 0 & \vec{r} \in B \end{cases}$



$P\{r, \theta, \phi\}$

$\vec{r} = (r, 0, 0)$

$\vec{\rho} = (\rho, \theta, \phi)$

$r_P = \sqrt{r^2 + \rho^2 - 2r\rho\cos\theta}$

show:

$$U = -\begin{cases} \frac{GM}{|r|} & r > R \\ \frac{GM}{2R^3}[3R^2 - r^2] & r \le R \end{cases}$$

use spherical coord for integration

$$U = \int_B \frac{G\sigma\,dV'}{|r-\rho|}$$

cartesian → spherical

$dV = \rho^2 \sin\theta\,d\theta\,d\phi\,d\rho$

$$U = \int_0^R \int_0^\pi \int_0^{2\pi} -\frac{G\sigma\rho^2\sin\theta}{\sqrt{r^2+\rho^2-2r\rho\cos\theta}}\,d\phi\,d\theta\,d\rho$$

$$U = -2\pi G\sigma \int_0^R \int_0^\pi \frac{\rho^2\sin\theta}{\sqrt{r^2+\rho^2-2r\rho\cos\theta}}\,d\theta\,d\rho$$

$u = r^2 + \rho^2 - 2r\rho\cos\theta$ $\theta = 0 \Rightarrow r^2+\rho^2-2r\rho = (r-\rho)^2$

$du = 2r\rho\sin\theta\,d\theta$ $\theta = \pi \Rightarrow r^2+\rho^2+2r\rho = (r+\rho)^2$

$$U = -2\pi G\sigma \int_0^R \rho^2 \int_{(r-\rho)^2}^{(r+\rho)^2} \frac{1}{2r\rho} u^{-1/2}\,du\,d\rho$$

$$= -2\pi G\sigma \int_0^R \rho^2 \left| \frac{1}{r\rho} u^{1/2} \right|_{(r-\rho)^2}^{(r+\rho)^2} d\rho$$

$$= -2\pi G\sigma \int_0^R \rho^2 \left| \frac{r+\rho}{r\rho} - \frac{|r-\rho|}{r\rho} \right| d\rho$$

$$= -2\pi G\sigma \int_0^R \rho\left(\frac{2\rho}{r}\right) d\rho \quad \Rightarrow \text{when } r > \rho$$

$$= -2\pi G\sigma \int_0^R \frac{2\rho^2}{r}\,d\rho$$

$$= -2\pi G\sigma\, \frac{2}{3}\frac{\rho^3}{r}\Big|_0^R$$

$U = -\frac{4}{3}\pi G\sigma \frac{R^3}{r}$ $M = \frac{4}{3}\pi\sigma R^3$

$$\boxed{U = -\frac{GM}{|r|} \Rightarrow r > R}$$

if ① $r > \rho$: $|r-\rho| = r - \rho$ $0 \le \rho \le r$

② $r < \rho$: $|r-\rho| = \rho - r$ $r \le \rho \le R$

① $U = -2\pi G\sigma \left\{ \int_0^r \rho^2 \left[\frac{r+\rho}{r\rho} - \frac{r-\rho}{r\rho}\right] d\rho + \int_r^R \rho^2 \left[\frac{r+\rho}{r\rho} - \frac{\rho-r}{r\rho}\right] d\rho \right.$

$U = -2\pi G\sigma \int_0^r \frac{2\rho^2}{r}\,d\rho + \int_r^R \rho^2 \frac{2}{\rho}\,d\rho$

$= -2\pi G\sigma \left[\int_0^r \frac{2\rho^2}{r}\,d\rho + \int_r^R 2\rho\,d\rho\right]$

$= -2\pi G\sigma \left[\frac{2}{3}\frac{\rho^3}{r}\Big|_0^r + \rho^2\Big|_r^R \right]$

$= -2\pi G\sigma \left[\frac{2}{3}r^2 + R^2 - r^2\right]$

$U = -2\pi G\sigma \left(R^2 - \frac{1}{3}r^2\right)$ $M = \frac{4\pi}{3}\sigma R^3$

$U = \frac{3}{2}\frac{GM}{R^3}\left(R^2 - \frac{1}{3}r^2\right)$

$$\boxed{U = -\frac{GM}{2R^3}(3R^2 - r^2) \quad r \le R}$$

13) given: above mass distr: $\frac{1}{2}\int_B U(r)\,dm$

13) given: above   mass distr: $\frac{1}{2}\int_B U(r)\,dm$

find: self potential $U_{11} = -\frac{3GM^2}{5R}$

$U = -\frac{GM}{2R^3}(3R^2 - r^2)$ $dm = \sigma\,dv$ $dv = r^2 \sin\theta\,dr\,d\theta\,d\phi$

$U_{11} = \frac{-1}{2}\int GM\left(\frac{3}{R} - \frac{r^2}{R^3}\right)\sigma\,dV'$

$= -\frac{1}{2}\int_0^R\int_0^\pi\int_0^{2\pi} \frac{GM}{R^3}(3R^2 - r^2)\,\sigma r^2 \sin\theta\,dr\,d\theta\,d\phi$

$= -\frac{\pi GM\sigma}{2R^3}\int_0^R\int_0^\pi (3R^2 - r^2) r^2 \sin\theta\,dr\,d\theta$

$= -\frac{\pi GM\sigma}{\cancel{2}R^3}\int_0^R (3R^2 - r^2) r^2 \left(-\cos\theta\Big|_0^\pi\right)^{(2)}$

$= -\pi\frac{GM\sigma}{R^3}\int_0^R (3R^2 - r^2) r^2\,dr$

$= -\frac{\pi GM\sigma}{R^3}\int_0^R (3R^2 r^2 - r^4)\,dr$

$= -\frac{\pi GM\sigma}{R^3}\left(R^2 r^3 - \frac{r^5}{5}\right)\Big|_0^R$

$= -\frac{\pi GM\sigma}{R^3}\left(R^5 - \frac{R^5}{5}\right)$

$= -\pi GM\sigma \frac{4}{5}R^2$

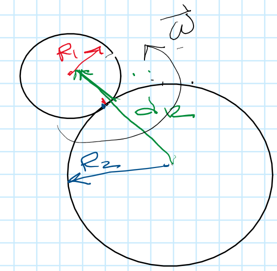
$= -\frac{4\pi}{5}GM\sigma R^2$ $\sigma = \frac{3M}{4\pi R^3}$

$= -\frac{3M^2 G R^2}{5R^3}$

$\boxed{U_{11} = -\frac{3GM^2}{5R}}$

14) given: Full 2BP sphere restricted w/ σ, R_1, R_2
resting on each other

$0 \le \frac{R_1}{R_2} \le 1$ $\quad q = \frac{R_1}{R_2}$



$m_i = \frac{4}{3}\pi\sigma R_i^3$ $\quad \frac{m_1}{m_2} = \frac{R_1^3}{R_2^3} = q^3$ $\quad M = m_1 + m_2 = m_2(1+q^3)$

$I_{sphere} = \frac{2}{5} m R^2$ $\quad r = R_1 + R_2$

$\vec{\Omega} = \vec{\omega}_1 = \vec{\omega}_2$

find: spin rate to fission $\quad E_{free}$ of system $\quad E_{free} = E_{tot} - (E_{11} + E_{22})$

plot radius ratio vs E_{free} $\quad \mu = \frac{m_1 m_2}{m_1 + m_2}$ $\quad \vec{\omega} = \hat{r}\,\Omega$

$E_{free} = E_{tot} - (U_{11} + U_{22})$

mutual potential: $U_{12} = -G\frac{m_1 m_2}{R_1 + R_2}$

self potentials: $U_{11} = -\frac{3}{5}\left(\frac{GM_1^2}{R_1}\right)$ $\quad U_{22} = -\frac{3}{5}\left(\frac{GM_2^2}{R_2}\right)$

Inertia:

$$\boxed{I_H = \frac{m_1 m_2}{m_1 + m_2} r^2 + \frac{2}{5}(m_1 R_1^2 + m_2 R_2^2)}$$

Energy:

$$T_{rot} = \frac{1}{2} I \Omega^2$$
$$= \frac{1}{2}(\mu r^2 + I_1 + I_2)\Omega^2$$

$$E_{tot} = T_{rot} + U_{12} + U_{11} + U_{22}$$

$$\boxed{E_{free} = \frac{1}{2}\Omega^2\left(\mu r^2 + \frac{2}{5}(m_1 R_1^2 + m_2 R_2^2)\right) - G\frac{m_1 m_2}{r}}$$

Angular momentum:

$$H_{tot} = H_{orbit} + H_{spin}$$

$$H_{tot} = \frac{m_1 m_2}{m_1 + m_2} r^2 \Omega + (I_1 + I_2)\Omega$$

$$\boxed{H_{rest} = (\mu r^2 + I_1 + I_2)\Omega} = \left(\mu r^2 + \frac{2}{5}(m_1 R_1^2 + m_2 R_2^2)\right)\Omega$$

Fission: spin = gravity

$\frac{\partial \mathcal{E}_H}{\partial r} = 0$ @ r

$$\mathcal{E}_H = \left[\frac{H^2}{2(\mu r^2 + I_s)} - \frac{G m_1 m_2}{r}\right] = 0$$

$$\frac{-H^2 \mu r}{(I_{tot})^2} + \frac{G m_1 m_2}{r^2} = 0$$

$$\frac{-H^2 \mu r}{(I_{tot})^2} + \frac{G m_1 m_2}{r^2} = 0$$

$$H^2 = \frac{G m_1 m_2 I_{tot}^2}{\mu_1 r^3}$$

$$\omega^2 = \frac{G(m_1 + m_2)}{r^3}$$

$$E_{free} = \frac{H^2}{2I_{tot}} - \frac{G m_1 m_2}{r} = \frac{1}{2} I_{tot} \omega^2 - \frac{G m_1 m_2}{r}$$